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In [3]: import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, classification_report

data = {
    'Age': ['youth', 'youth', 'middle_aged', 'senior', 'senior', 'senior', 'middle_aged',
    'Income': ['high', 'high', 'high', 'medium', 'low', 'low', 'low', 'medium', 'low', 'medi
    'Student': ['no', 'no', 'no', 'no', 'yes', 'yes', 'yes', 'no', 'yes', 'yes', 'yes', 'no', '
    'Credit_Rating': ['fair', 'excellent', 'fair', 'fair', 'fair', 'excellent', 'excellen
    'Buys_Computer': ['no', 'no', 'yes', 'yes', 'yes', 'no', 'yes', 'no', 'yes', 'yes', 'yes'
}

df = pd.DataFrame(data)

print("Original Dataset:\n")
print(df)

le = LabelEncoder()

for column in df.columns:
    df[column] = le.fit_transform(df[column])

print("\nEncoded Dataset:\n")
print(df)

X = df.drop('Buys_Computer', axis=1)
y = df['Buys_Computer']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = DecisionTreeClassifier(criterion='entropy') # using entropy (ID3 concept)

model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("\nAccuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
plt.figure(figsize=(12,8))
plot_tree(
    model,
    feature_names=X.columns,
    class_names=['No', 'Yes'],
    filled=True
)
plt.title("Decision Tree - Buys Computer Dataset")
plt.show()

```

Original Dataset:

	Age	Income	Student	Credit_Rating	Buys_Computer
0	youth	high	no	fair	no
1	youth	high	no	excellent	no
2	middle_aged	high	no	fair	yes
3	senior	medium	no	fair	yes
4	senior	low	yes	fair	yes
5	senior	low	yes	excellent	no
6	middle_aged	low	yes	excellent	yes
7	youth	medium	no	fair	no
8	youth	low	yes	fair	yes
9	senior	medium	yes	fair	yes
10	youth	medium	yes	excellent	yes
11	middle_aged	medium	no	excellent	yes
12	middle_aged	high	yes	fair	yes
13	senior	medium	no	excellent	no

Encoded Dataset:

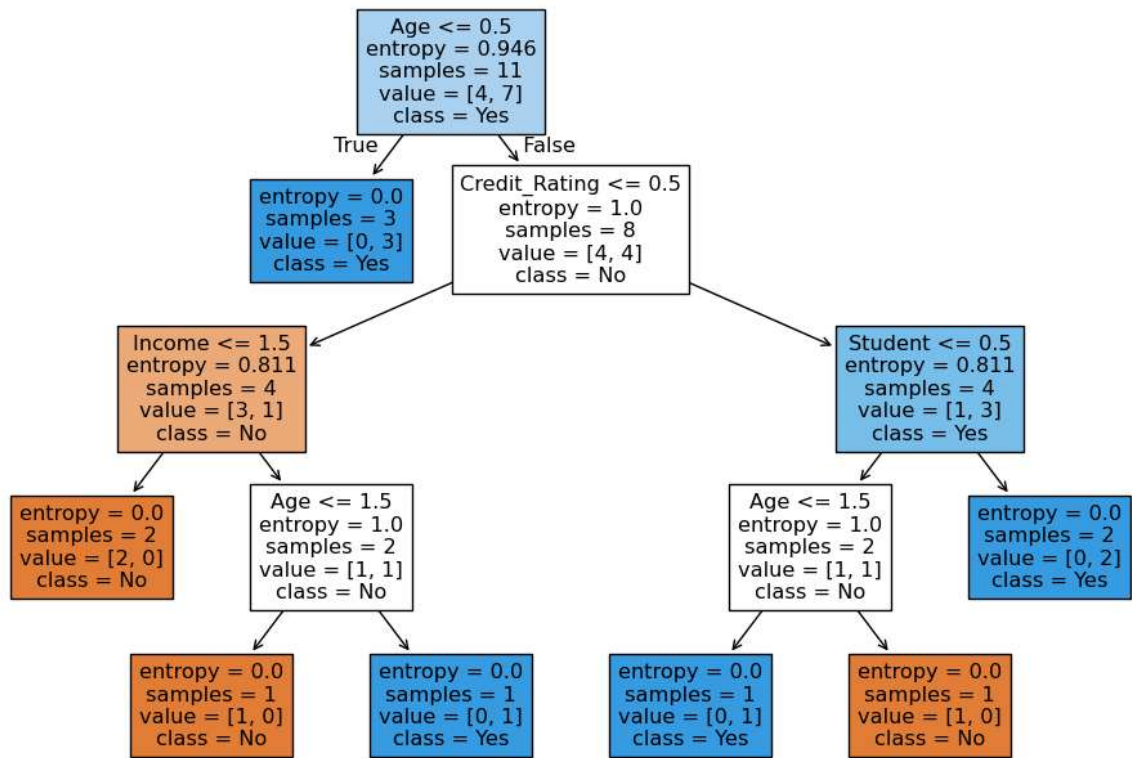
	Age	Income	Student	Credit_Rating	Buys_Computer
0	2	0	0	1	0
1	2	0	0	0	0
2	0	0	0	1	1
3	1	2	0	1	1
4	1	1	1	1	1
5	1	1	1	0	0
6	0	1	1	0	1
7	2	2	0	1	0
8	2	1	1	1	1
9	1	2	1	1	1
10	2	2	1	0	1
11	0	2	0	0	1
12	0	0	1	1	1
13	1	2	0	0	0

Accuracy: 1.0

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	1
1	1.00	1.00	1.00	2
accuracy			1.00	3
macro avg	1.00	1.00	1.00	3
weighted avg	1.00	1.00	1.00	3

Decision Tree - Buys Computer Dataset



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